# Tenglamalarni yechish

Noma'lum miqdorni harf bilan belgilab, masalani tenglama bilan yechamiz.

1-QISM

## Mavzu

ESLAB OLAYLIK

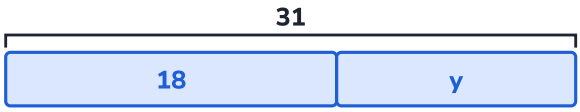
- O'xshash hadlarni qo'shamiz va ayiramiz: $x + 4x = 5x$, $6y - y = 5y$, $3a + 2 + a = 4a + 2$.
- Noma'lum qo'shiluvchi = yig'indi − ma'lum qo'shiluvchi; noma'lum ko'paytuvchi = ko'paytma : ma'lum ko'paytuvchi.
- Noma'lum kamayuvchi = ayirma + ayriluvchi; noma'lum bo'linuvchi = bo'linma · bo'luvchi.

1-QOIDA · ALGEBRAIK TENGLAMA

$3x + 5 = 26$ – **algebraik tenglama**. $x$ ning qiymatini topish tenglamani $x$ ga nisbatan yechishni anglatadi.

1-MISOL · KITOB JAVONI

Ikki tokchada jami 31 ta kitob bor. Pastki tokchada 18 ta kitob. Yuqori tokchada nechta kitob bor?




1. Yuqori tokchadagi kitoblar sonini $y$ bilan belgilaymiz: $18 + y = 31$.
2. $y = 31 - 18 = 13$. Yuqori tokchada $13$ ta kitob bor.

2-MISOL · PISHIRIQLAR

Har bir qutiga 6 tadan pishiriq joylanadi. 42 ta pishiriq uchun nechta quti kerak?

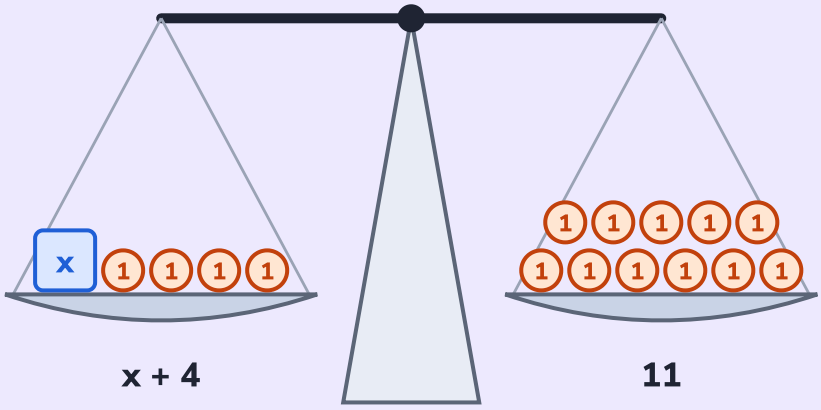
1. Qutilar sonini $n$ bilan belgilaymiz: $6n = 42$.
2. $n = 42 : 6 = 7$. $7$ ta quti kerak.

Ikki palladan bir xil tosh olsam, tarozi qiyshaymaydimi?



**USUL · PALLALI TAROZI**

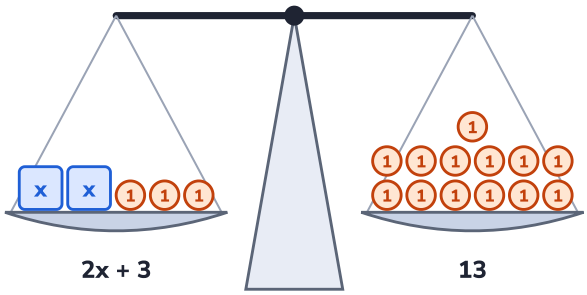
Har bir doiracha — 1 ga teng birlik tosh, $x$ — noma'lum og'irlikdagi quti. Tarozi muvozanatda: $x + 4 = 11$.




1. Tarozining chap pallasidan 4 ta birlik toshni olib tashlaymiz: $x + 4 - 4$.
2. O'ng pallasidan ham 4 ta birlik toshni olib tashlaymiz: $11 - 4$.
3. $x$ ning qiymatini topamiz: $x = 7$. Bu son — tenglamaning **yechimi**: harf o'rniga qo'yilganda tenglik to'g'ri bo'ladigan son. Tekshiramiz: $7 + 4 = 11$.

**1-MISOL · IKKI QUTI**

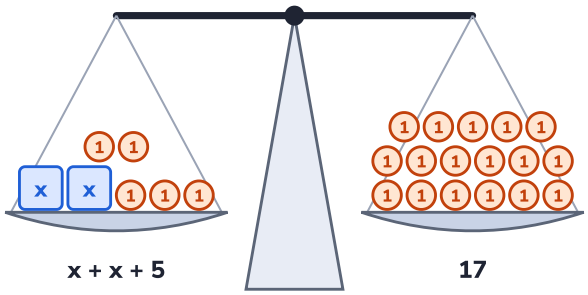
Tarozidagi tenglamani yeching.




1. Ikkala palladan 3 tadan tosh olamiz: $2x = 13 - 3 = 10$.
2. Ikki quti 10 ta toshga teng, bitta quti — $10 : 2$: $x = 5$.

**2-MISOL · AVVAL SODDALASHTIRAMIZ**

$x + x + 5 = 17$ tenglamani yeching.




1. Qutilarni birlashtiramiz: $x + x = 2x$, ya'ni $2x + 5 = 17$.
2. $2x = 17 - 5 = 12$.
3. $x = 12 : 2 = 6$.

**USUL · BO'LAKLAR VA TENGLAMA**

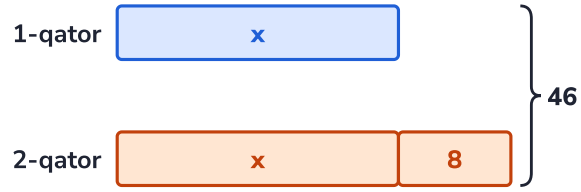



2 bo'lak + 10 = 36 ni $2x + 10 = 36$ deb oling va uni $x$ ga nisbatan yeching.

| Bo'laklar bilan | Tenglama bilan |
|---|---|
| 2 bo'lak + 10 = 36 | $2x + 10 = 36$ |
| 2 bo'lak = 36 − 10 = 26 | $2x = 36 - 10 = 26$ |
| 1 bo'lak = 26 : 2 = 13 | $x = 26 : 2 = 13$ |

**1-MISOL · KO'CHATLAR**

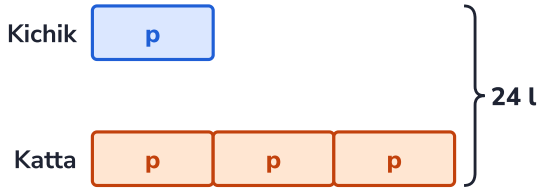
Ikki qatorga jami 46 tup ko'chat ekildi. Ikkinchi qatorda birinchisidan 8 tup ko'p. Ikkinchi qatorda necha tup ko'chat bor?




1. 1-qator $x$, 2-qator $x+8$: $x+x+8=46$, ya'ni $2x+8=46$.
2. $2x=46-8=38$, $x=38:2=19$.
3. Ikkinchi qatorda $19+8=27$ tup.

**2-MISOL · IDISHLAR**

Katta idishga kichik idishdagidan 3 marta ko'p suv sig'adi. Ikkalasiga birga 24 l suv sig'adi. Har biriga necha litr suv sig'adi?




1. Kichik idish $p$, katta idish $3p$: $p+3p=24$, ya'ni $4p=24$.
2. $p=24:4=6$. Kichik idishga $6$ l, kattasiga $3\cdot 6=18$ l.



Miqdorlar uchta bo'lsa ham bitta harf yetadimi?

**USUL · UCHTA MIQDOR**

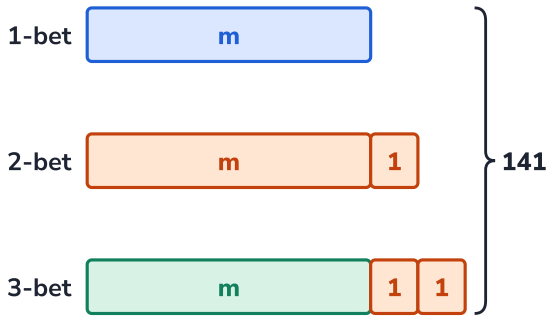
Uchta son, har biri oldingisidan 2 ta ko'p, yig'indisi 36. $x$ lar alohida, sonlar alohida yig'iladi:

$$x+(x+2)+(x+4)=36 \to 3x+6=36 \to 3x=36-6=30 \to x=10.$$

Tekshirish: $10+12+14=36$.

**1-MISOL · KITOB BETLARI**

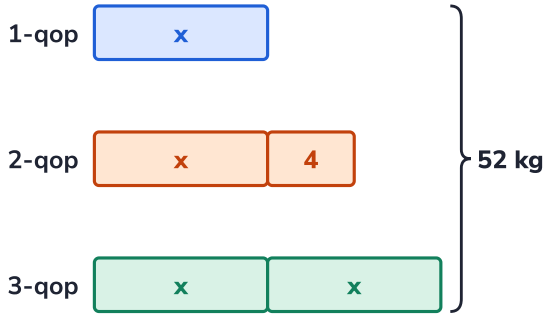
Aziz kitobning ketma-ket uchta betini o'qidi. Bu bet sonlarining yig'indisi 141. Aziz qaysi betlarni o'qidi?




1. Betlar: $m$, $m+1$, $m+2$. Uchta $m$ bor, ortiqcha 1 va 2 ni ayiramiz: $3m = 141 - 1 - 2 = 138$.
2. $m = 138 : 3 = 46$.
3. Betlar: $46, 47, 48$. Tekshirish: $46 + 47 + 48 = 141$.

**2-MISOL · GURUCH QOPLARI**

Uchta qopda jami 52 kg guruch bor. Ikkinchi qopda birinchisidan 4 kg ko'p, uchinchi qopda birinchisidan 2 marta ko'p guruch bor. Har bir qopda necha kg guruch bor?




1. $x + (x+4) + 2x = 52$, ya'ni $4x + 4 = 52$.
2. $4x = 52 - 4 = 48$, $x = 48 : 4 = 12$.
3. Qoplarda: $12$ kg, $16$ kg, $24$ kg. Tekshirish: $12 + 16 + 24 = 52$.

2-QISM

# Misollar ishlash

### 1-MISOL · OTA VA O'G'IL

Hozir ota 38 yoshda, o'g'li 10 yoshda. Necha yildan keyin ularning yoshlari yig'indisi 60 bo'ladi?

1. Noma'lum yillar sonini $n$ bilan belgilaymiz.
2. $n$ yildan keyin o'g'il $10 + n$, ota $38 + n$ yoshda bo'ladi.
3. Tenglama tuzamiz: $(10 + n) + (38 + n) = 60$.
4. Sonlarni va harflarni yig'amiz: $2n + 48 = 60$.
5. $2n = 60 - 48 = 12$.
6. $n = 12 : 2 = 6$. Javob: $6$ yildan keyin.
7. Tekshiramiz: o'g'il 16, ota 44 yoshda bo'ladi; $16 + 44 = 60$.

### 2-MISOL · TO'G'RI TO'RTBURCHAK

To'g'ri to'rtburchakning bo'yi enidan 4 cm uzun. Uning perimetri 40 cm. To'g'ri to'rtburchakning eni va bo'yini toping.

1. Enini $x$ bilan belgilaymiz. Bo'yi $x + 4$.
2. Perimetr — to'rtta tomon yig'indisi: $x + (x + 4) + x + (x + 4) = 40$.
3. Soddalashtiramiz: $4x + 8 = 40$.
4. $4x = 40 - 8 = 32$.
5. $x = 32 : 4 = 8$. Eni $8$ cm, bo'yi $8 + 4 = 12$ cm.
6. Tekshiramiz: $8 + 12 + 8 + 12 = 40$.

3-MISOL · KO'CHAT EKISH

Uchta sinf jami 63 tup ko'chat ekdi. 6-B sinf 6-A sinfdan 2 marta ko'p, 6-V sinf esa 6-A sinfdan 5 tup kam ekdi. Har bir sinf necha tup ko'chat ekdi?

1. 6-A sinf $x$, 6-B sinf $2x$, 6-V sinf $x - 5$ tup ekdi.
2. Tenglama: $x + 2x + (x - 5) = 63$.
3. Harflarni yig'amiz: $4x - 5 = 63$.
4. $4x$ — kamayuvchi, uni topish uchun 5 ni qo'shamiz: $4x = 63 + 5 = 68$.
5. $x = 68 : 4 = 17$.
6. 6-A: $17$ tup, 6-B: $2 \cdot 17 = 34$ tup, 6-V: $17 - 5 = 12$ tup.
7. Tekshiramiz: $17 + 34 + 12 = 63$.

Men bir marta «ikkinchi qatorda nechta?» savoliga $x$ ning o'zini yozib qo'yganman.



DIQQAT!

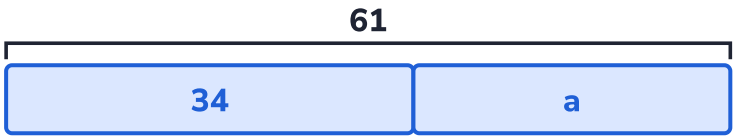
1. $3x + 5 = 26$ da ✗ $x = 26 : 3 - 5$ deb yozish xato. Avval 5 ni olib tashlang: ✓ $3x = 21, x = 7$
2. «8 tup ko'p» bo'lsa, ikkala qator ham yoziladi: ✗ $x + 8 = 46$ → ✓ $x + (x + 8) = 46$
3. Ayirish bo'lsa, sonni qo'shamiz: ✗ $4x - 5 = 63, 4x = 58$ → ✓ $4x = 63 + 5 = 68$
4. Yillar ikkala kishiga qo'shiladi: ✗ $(10 + n) + 38 = 60$ → ✓ $(10 + n) + (38 + n) = 60$
5. So'ralgan miqdorni javob qilamiz. «Ikkinchi qatorda nechta?»: ✗ $x = 19$ → ✓ $x + 8 = 27$ **tup**

3-QISM

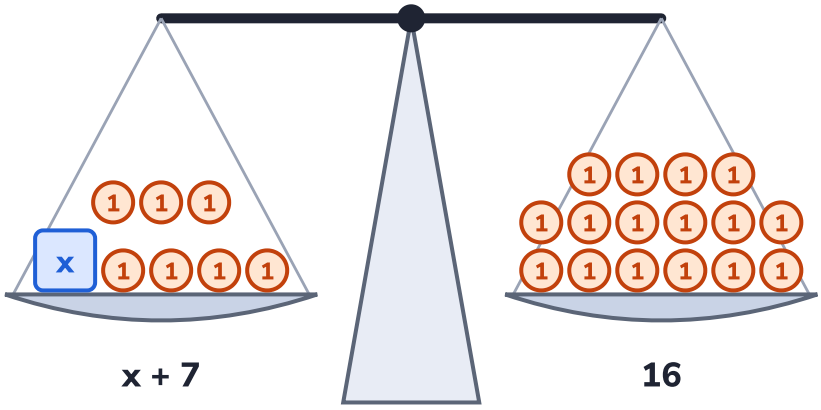
# Mustahkamlash

1. Noma'lum sonni toping: a) $y + 19 = 47$; b) $8k = 96$; c) $25 + m = 60$; d) $11n = 77$ .

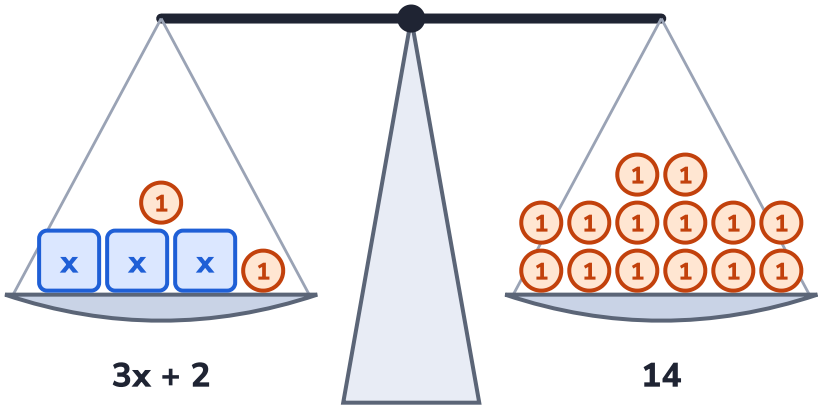
2. Chizmali modelga mos tenglama tuzing va uni yeching.




3. Tarozi muvozanatda. Bitta quti nechta birlik toshga teng?




4. Tarozi rasmiga mos tenglama tuzing va $x$ ni toping.



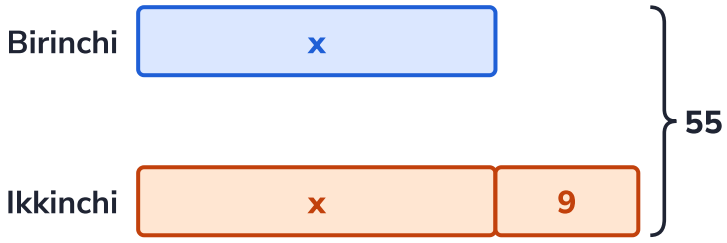

5. Yeching: a) $4x + 9 = 41$; b) $6y + 15 = 51$; c) $x + 3x = 36$; d) $2p + p + 7 = 37$.

6. Birinchi qutida $x$ ta kubik bor. Ikkinchi qutida undan 6 ta ko'p, uchinchi qutida birinchisidan 3 marta ko'p kubik bor. a) Har bir qutidagi kubiklar sonini va jami sonini $x$ orqali yozing. b) Jami 46 ta kubik bo'lsa, har bir qutida nechta kubik bor?

7. **Xatoni toping.** Sherzod $2x - 6 = 18$ tenglamani shunday yechdi: $2x = 12$, $x = 6$. Xato qayerda? To'g'ri yeching.

8. $x$, $a$ va $m$ ni toping: a) $5x - 4 = 31$; b) $3a + 2a - 6 = 44$; c) $m + (m + 1) + (m + 2) = 48$.

9. Chizmali modelga mos tenglama tuzing. $x$ ni va ikkinchi qatordagi sonni toping.




10. Teng yonli uchburchakning ikkita teng tomoni $x$ cm dan, uchinchi tomoni 7 cm. Perimetri 31 cm. $x$ ni toping.

11. Sehrli kvadratda har bir qator, ustun yoki diagonaldagi sonlar yig'indisi teng va har bir son faqat bir marta ishlatilgan. Markazdagi son $x$.

| | | |
|---|---|---|
| $x - 1$ | ? | $x + 3$ |
| ? | $x$ | $x - 4$ |
| $x - 3$ | $x + 2$ | ? |

a) Diagonal $(x + 3) + x + (x - 3)$ ning yig'indisini toping va uchta bo'sh katakni to'ldiring. b) Chap ustundagi sonlar yig'indisi 27 bo'lsa, $x$ nechaga teng? c) Shu $x$ uchun kvadratdagi sonlarni yozing.

## Matnli masalalar

1. Ikki qavatli avtobusda 57 yo'lovchi bor. Pastki qavatda yuqori qavatdagidan 9 yo'lovchi kam. Yuqori qavatda nechta yo'lovchi bor?

2. Sayohatchi uch kunda 75 km yo'l bosdi. Birinchi kuni $x$ km, ikkinchi kuni undan 2 marta ko'p, uchinchi kuni 15 km yurdi. Ikkinchi kuni necha km yurgan?

3. Hozir aka 15 yoshda, uka 9 yoshda. Necha yildan keyin ularning yoshlari yig'indisi 40 bo'ladi?

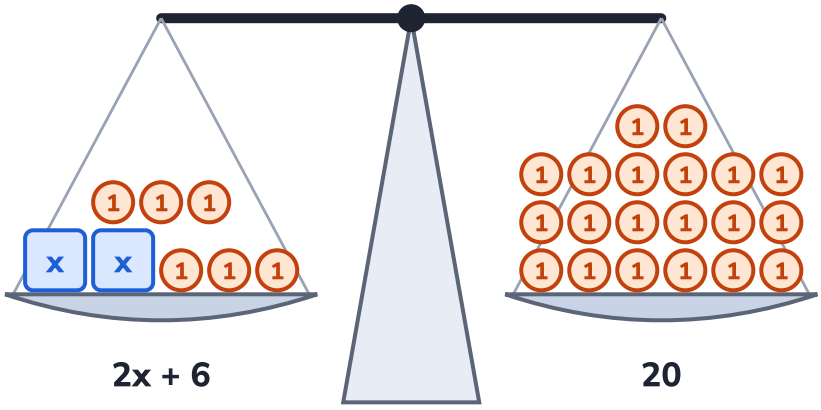
4. To'g'ri to'rtburchakning bo'yi enidan 3 marta uzun. Perimetri 64 cm. To'g'ri to'rtburchakning yuzini toping.

5. Bog'bon 4 ta savatni bir xil miqdorda uzum bilan to'ldirdi. Bozorga 18 kg uzum olib ketdi, uyda 30 kg uzum qoldi. Bitta savatda necha kg uzum bo'lgan?

4-QISM

# Mustaqil ish

1. Yeching va javobni tenglamaga qo'yib tekshiring: a) $8x + 13 = 69$; b) $7y - 9 = 47$.
2. Ikkala palladan nechta tosh olib tashlaysiz? Bitta quti nechta toshga teng?




3. Kutubxonaga ketma-ket uch kunda jami 87 kitobxon keldi. Har kuni oldingi kundagidan 5 kishi ko'p keldi. Birinchi kuni nechta kitobxon kelgan?
4. 50 cm simdan tomoni $x$ cm bo'lgan kvadrat yasaldi va 6 cm sim ortib qoldi. Kvadratning tomoni necha santimetr?
5. Hozir ona 41 yoshda, qizi 13 yoshda. Necha yildan keyin ularning yoshlari yig'indisi 74 bo'ladi?
6. Uchburchakning ikki tomoni teng, uchinchi tomoni ulardan 3 cm qisqa. Perimetri 36 cm. Uchburchakning tomonlarini toping.

5-QISM

# Javoblar

## Mustahkamlash

**1-masala**

a) $y = 47 - 19 = 28$; b) $k = 96 : 8 = 12$; c) $m = 60 - 25 = 35$; d) $n = 77 : 11 = 7$.

**2-masala**

$34 + a = 61$, $a = 61 - 34 = 27$.

**3-masala**

$x + 7 = 16$, $x = 16 - 7 = 9$.

**4-masala**

$3x + 2 = 14$, $3x = 12$, $x = 4$.

**5-masala**

a) $4x = 32$, $x = 8$; b) $6y = 36$, $y = 6$; c) $4x = 36$, $x = 9$; d) $3p + 7 = 37$, $3p = 30$, $p = 10$.

**6-masala**

a) $x$, $x + 6$, $3x$; jami $5x + 6$. b) $5x + 6 = 46$, $5x = 40$, $x = 8$. Qutilarda: 8, 14 va 24 ta.

**7-masala**

6 ni ayirish emas, qo'shish kerak edi: $2x = 18 + 6 = 24$, $x = 12$. Tekshirish: $24 - 6 = 18$.

**8-masala**

a) $5x = 35$, $x = 7$; b) $5a - 6 = 44$, $5a = 50$, $a = 10$; c) $3m + 3 = 48$, $3m = 45$, $m = 15$.

**9-masala**

$x + (x + 9) = 55$, $2x + 9 = 55$, $2x = 46$, $x = 23$. Ikkinchi qatorda $23 + 9 = 32$.

**10-masala**

$2x + 7 = 31$, $2x = 24$, $x = 12$.

**11-masala**

a) $(x + 3) + x + (x - 3) = 3x$. Bo'sh kataklar: yuqori o'rta $3x - (x - 1) - (x + 3) = x - 2$, o'rta chap $3x - x - (x - 4) = x + 4$, pastki o'ng $3x - (x - 3) - (x + 2) = x + 1$. b) $(x - 1) + (x + 4) + (x - 3) = 3x = 27$, $x = 9$. c) 8, 7, 12; 13, 9, 5; 6, 11, 10.

## Matnli masalalar

**1-masala**

Pastki qavat $x$, yuqori $x + 9$. $2x + 9 = 57$, $2x = 48$, $x = 24$. Yuqori qavatda $24 + 9 = 33$ yo'lovchi.

**2-masala**

$x + 2x + 15 = 75$, $3x = 60$, $x = 20$. Ikkinchi kuni $2 \cdot 20 = 40$ km.

**3-masala**

$(15 + n) + (9 + n) = 40$, $2n + 24 = 40$, $2n = 16$, $n = 8$. 8 yildan keyin (23 va 17 yosh).

**4-masala**

Eni $x$, bo'yi $3x$: $x + 3x + x + 3x = 64$, $8x = 64$, $x = 8$. Bo'yi 24 cm, yuzi $24 \cdot 8 = 192$ cm².

**5-masala**

$4x - 18 = 30$, $4x = 48$, $x = 12$ kg.

## Mustaqil ish

**1-masala**

a) $8x = 56$, $x = 7$; tekshirish: $56 + 13 = 69$. b) $7y = 56$, $y = 8$; tekshirish: $56 - 9 = 47$.

**2-masala**

$2x + 6 = 20$, $2x = 14$, $x = 7$.

**3-masala**

$x + (x + 5) + (x + 10) = 87$, $3x + 15 = 87$, $3x = 72$, $x = 24$ kitobxon (keyin 29 va 34).

**4-masala**

$4x + 6 = 50$, $4x = 44$, $x = 11$ cm.

**5-masala**

$(41 + n) + (13 + n) = 74$, $2n + 54 = 74$, $2n = 20$, $n = 10$. 10 yildan keyin (51 va 23 yosh).

**6-masala**

Teng tomonlar $x$ dan, uchinchisi $x - 3$: $x + x + (x - 3) = 36$, $3x - 3 = 36$, $3x = 39$, $x = 13$. Tomonlari 13 cm, 13 cm, 10 cm.